Lab 7: Metabolism — Enzymes, ATP, and Energy Flow

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By the end of this lab, you will be able to:

1. **Explain** how enzymes lower activation energy without being consumed.
2. **Calculate** reaction rates and compare an uncatalyzed reaction to an enzyme-catalyzed one.
3. **Model** enzyme activity under normal, high-heat, low-pH, and inhibited conditions using stated rates.
4. **Predict** how temperature, pH, and inhibitors change enzyme shape and function.
5. **Model** active-site specificity with a paper shape game, and distinguish a blocked active site from a denatured one.
6. **Model ATP** as the cell's short-term energy-transfer molecule.
7. **Connect** metabolism to photosynthesis and cellular respiration as linked pathways.

Overview

Metabolism is the sum of all chemical reactions in a cell. Most of those reactions are too slow on their own, so cells depend on **enzymes** — proteins that speed reactions by lowering **activation energy**, the energy needed to get a reaction started.

Today you will run a **paper-and-pencil enzyme simulation**, calculating how many products an enzyme makes each minute under four conditions. You will then play a hands-on **active site shape game** with torn paper scraps to see exactly *why* shape matters, work through the ATP energy cycle, and map how energy flows from sunlight through photosynthesis, respiration, and ATP to cellular work. Everything is paper and pencil — no equipment.

Materials Needed

- Pencil or pen
- Calculator (optional, for division)
- This worksheet
- A few spare paper scraps for the shape game

Key Terms

Term	Definition
Metabolism	All chemical reactions in a cell or organism
Activation energy	Energy required to start a reaction
Enzyme	Protein catalyst that speeds up a reaction without being consumed
Substrate	Reactant that binds to an enzyme's active site
Active site	The region of an enzyme where the substrate fits
Competitive inhibitor	Molecule that blocks the active site without destroying the enzyme
Denaturation	Loss of an enzyme's shape, which disrupts its function
ATP	Cellular energy-coupling molecule (adenosine triphosphate)

Part 1: Activation Energy and Baseline Rates

Without an enzyme, most reactions happen slowly because substrates rarely collide with enough energy to react. An enzyme's active site holds substrates in just the right position, lowering the activation energy barrier — but the enzyme itself is **not used up**.

> **1.** In one sentence, what does an enzyme do to activation energy, and why does that make the reaction faster?

2. A reaction without an enzyme produces **2 products per minute**. The same reaction with its enzyme produces **18 products per minute**. Show your work: how many times faster is the enzyme condition?

Part 2: Enzyme Rate Simulation

How the Simulation Works

Your enzyme, **Sucrase-X**, breaks one substrate molecule into products. One "minute" is one round of the simulation. Each condition below has a **stated rate** (products made per minute). Using the rules in Table 1, calculate the **cumulative total** of products at the end of each minute and record it in Table 2. Then compute the average rate per minute for the full 3 minutes.

> ***Example:*** *If a condition made 4 products in minute 1 and 4 in minute 2, the cumulative totals would be* ***4*** *(after 1 min) and* ***8*** *(after 2 min).*

Table 1 — Simulation Rules (given)

Condition	What happens to the enzyme	Min 1	Min 2	Min 3
Normal (37 °C, pH 7)	Active site keeps its shape	12	12	12
High heat (80 °C)	Works at first, then denatures	12	3	0
Low pH (pH 2)	Acid warps the active site, but does not destroy it	6	6	6
Competitive inhibitor	Inhibitor occupies the active site ⅔ of the time	4	4	4

(Rates are products made per minute.)

Table 2 — Your Results (fill in)

Condition	Products after 1 min	Products after 2 min	Products after 3 min	Average products/min
Normal (37 °C, pH 7)				
High heat (80 °C)				
Low pH (pH 2)				
Competitive inhibitor				

Check yourself: the Normal condition should total 36 products after 3 minutes.

3. Which condition caused the largest drop in enzyme activity? Use numbers from your table to support your answer.

4. Explain why a heat-denatured enzyme cannot simply keep working at the same rate. Refer to the **shape of the active site** in your answer.

5. The low-pH condition slowed the enzyme but never stopped it completely, while high heat stopped it by minute 3. Based on that pattern, which condition do you predict is **reversible** if the cell restores normal pH, and which is likely **permanent**? Why?

6. A competitive inhibitor slows the enzyme by blocking the active site, but the enzyme's shape is still intact. Predict what would happen to the rate if the cell added **much more substrate**, and explain your reasoning.

Part 3: Active Site Shape Game

This game shows **why** the numbers in Table 1 came out the way they did. Everything is made from torn paper — no cutting tools, no fasteners.

Step 1 — Make your enzymes. On paper, draw **two** active-site shapes, each inside its own box outlined with a **dashed border** (the dashed border is your tear line — tear slowly, corner to corner):

- **Normal enzyme:** a wide "U" notch, about 4 cm across and 3 cm deep.
- **Denatured enzyme:** the same "U" notch, but **warped** — bend one arm of the U outward so it no longer matches the substrate shape.

Step 2 — Make your substrates and hypothesis. Tear out **7 small paper scraps**. Label **4** of them **S** (substrate) and **3** of them **I** (inhibitor decoy). Tear slowly and steadily, corner to corner.

7. Write your hypothesis BEFORE testing: which scraps will fit which enzyme, and what will the inhibitor decoys do?

Step 3 — Test-fit. Press each scrap into each enzyme notch and record whether it fits:

- **Normal enzyme: S** scraps fit snugly into the notch. **I** scraps also fit — but they **block the site**. A stuck inhibitor means the enzyme is **occupied, not denatured**: the shape is still fine; something else is sitting in the doorway.
- **Denatured enzyme: nothing fits**, because the warped notch no longer matches the substrate's shape.

Step 4 — Tally your results.

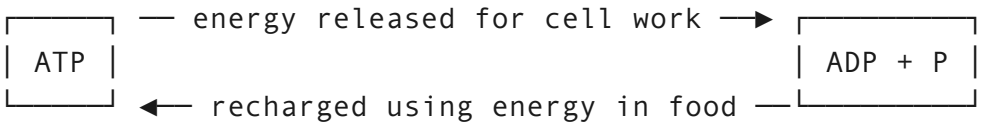
Scrap	Fits normal enzyme	Fits denatured enzyme
Substrate S (4 scraps)		
Inhibitor I (3 scraps)		

8. Compare your results to your hypothesis. Did anything surprise you? Explain what the **stuck inhibitor** and the **warped notch** each did to the enzyme, using the words **shape** and **active site**.

9. Connect back to Table 1: which simulated condition does your **denatured** enzyme model — high heat or low pH? Which condition does the **stuck inhibitor** model? Explain both choices in terms of what your scraps showed you.

Part 4: ATP — The Cell's Energy Currency

Cells cannot use a single large burst of energy from food directly. Instead, respiration transfers small packets of energy into **ATP**. When the cell needs to do work — building proteins, moving materials, contracting muscle — it breaks off ATP's third phosphate. The ATP–ADP cycle below runs continuously:



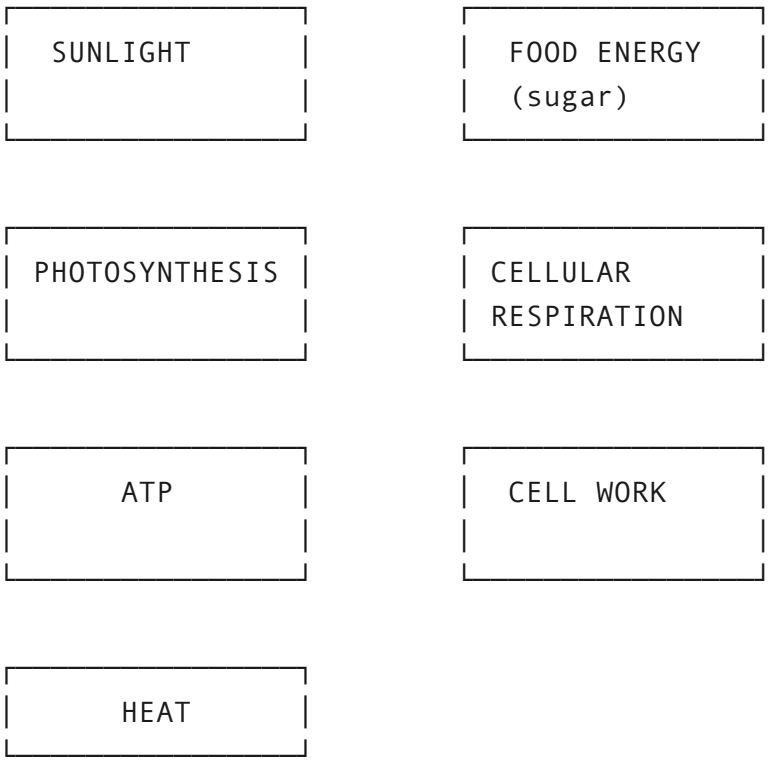
10. Fill in the blanks, using the cycle diagram above for the first two. ATP releases energy when it loses its third _ ***group, becoming*** _. The cell recharges it using energy harvested from _ _.

11. ATP is often called the cell's "energy currency." Explain what that analogy means, and give **one way** it can be misleading.

Part 5: Metabolism Map — Following the Energy

Photosynthesis and cellular respiration are linked: photosynthesis **builds** sugar using light energy, and respiration **breaks down** fuel to harvest that energy as ATP. Energy enters living systems as sunlight and leaves as heat; matter (carbon, oxygen, hydrogen) cycles between the two processes.

12. On the **printed map below**, draw arrows **directly on the paper** connecting the terms, so that every arrow you draw traces a real flow of **matter or energy**. Write a short label (for example: "light energy," "sugar," "ATP," "heat") on each arrow. Then, in the answer box, list **each arrow you drew** in the form `FROM -> TO: label (matter or energy)` — one line per arrow, so your on-paper arrows and your list match exactly.



13. Using your map as evidence, explain why metabolism is not just "breaking down food," but a coordinated **network** of building and breaking reactions.

Exit Check

Before you turn in, confirm you can do each item from the Learning Objectives without your notes:

- State what an enzyme does to activation energy in one sentence.
- Use the Table 1 rates to explain why 80 °C stopped the reaction but pH 2 only slowed it.
- Use your shape-game results to explain the difference between a **competitive inhibitor** (blocks the active site, enzyme intact) and **denaturation** (active site shape destroyed).
- Recite the ATP ↔ ADP + P cycle from memory.
- Trace one full energy pathway from sunlight to cell work on your map.